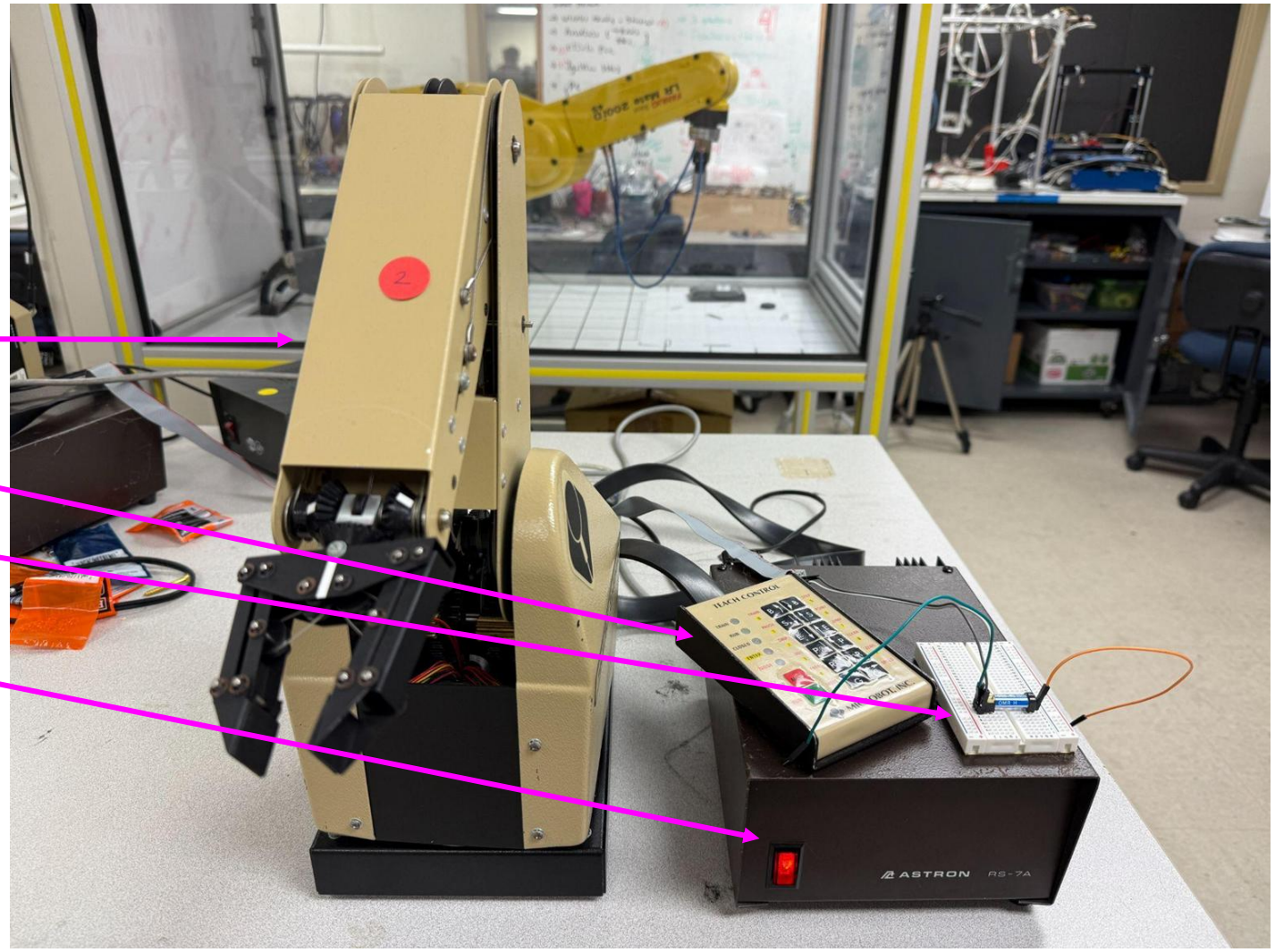
MMET 410 – Lab 10

Lab report requirements

- Task 1 - Describe what does each line of the program
- Task 2 - Test the relay. Answer the questions asked in Task 2.
- Task 3 and 4 - Draw the relay connection shown in figure 2, with PLC (slot 4, IB16) and cable from the robot. Connect them in PLC

Report due tomorrow midnight.

1. Microbot
2. Teaching pendant
3. Breadboard + small relay
4. Power Supply

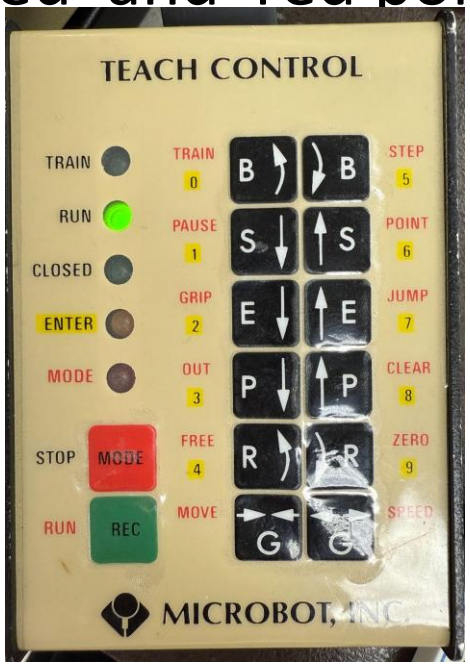


Task 1 Check Microbot TechMover

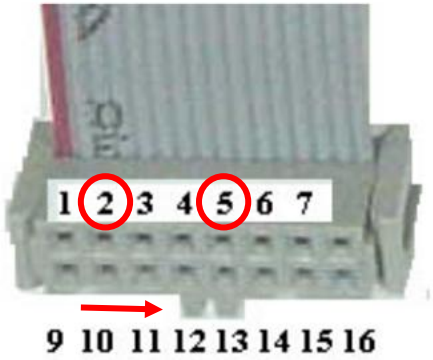


1. Be careful to confirm the connection between red and red point
2. Programming the Microbot

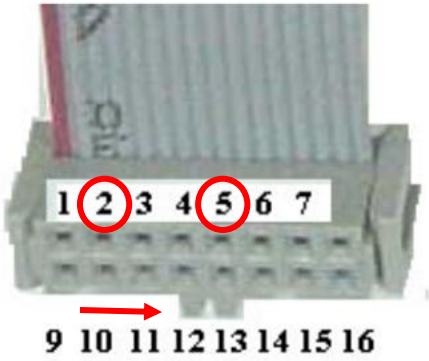
<Mode><Clear> at the same time (to clear memory)
<Mode>OUT 5<Mode> 1
<Mode>PAUSE 2
<Mode>OUT 5<Mode> 0
<Mode>PAUSE 2
<Mode>
<Run>



3. In your report:
   Describe what each line of the program does.
4. Check the output voltage (pin2 and pin5).



Task 2 Test relay functionality using output from Microbot TeachMover



1.Test relay and answer the question.



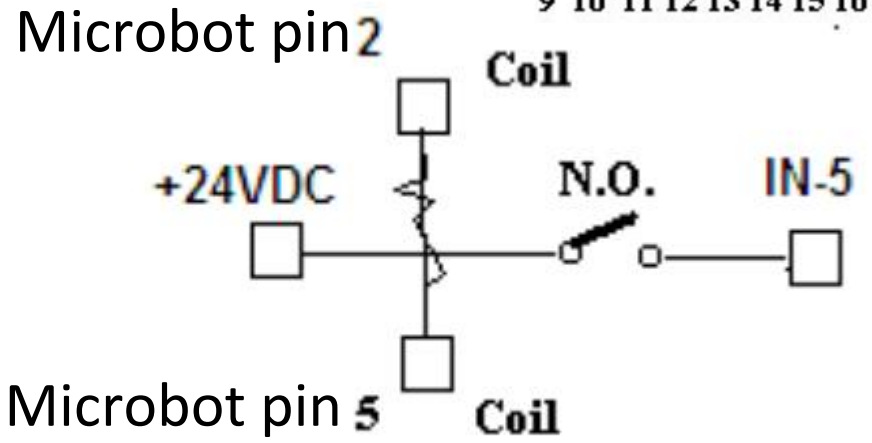


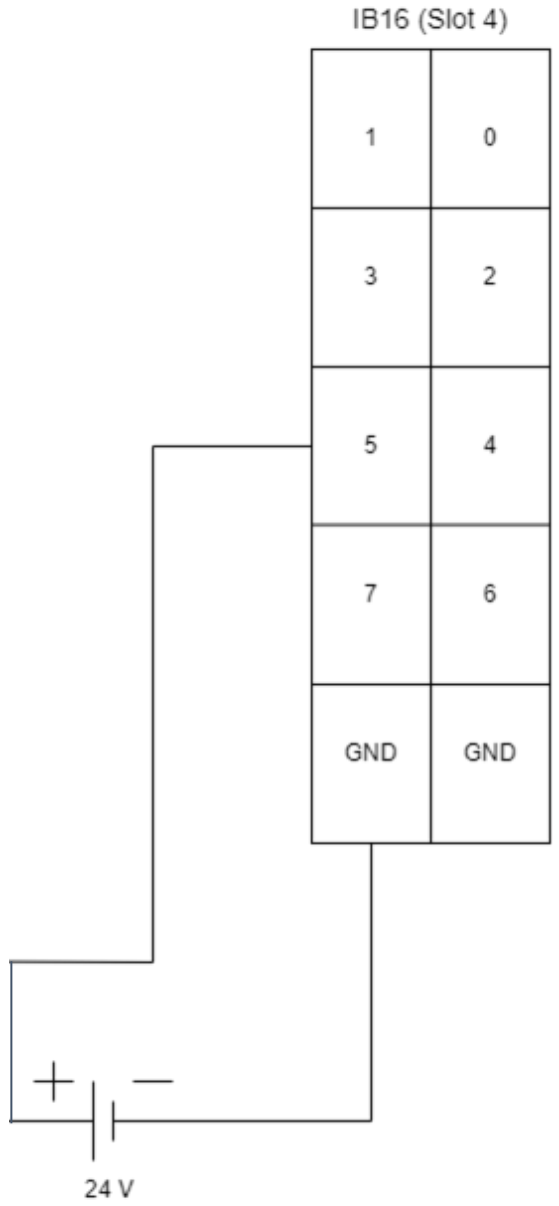
Figure 2. Schematic for relay.

2.In your report: Answers to the questions.

Task 3 DC input module at slot 4

1.Use either 24 VDC from the big relay machine
Or Get another power supply that provide 24VDC.

2.In your report:
Answers to the questions and
Drawing of the wire connection



Task 4

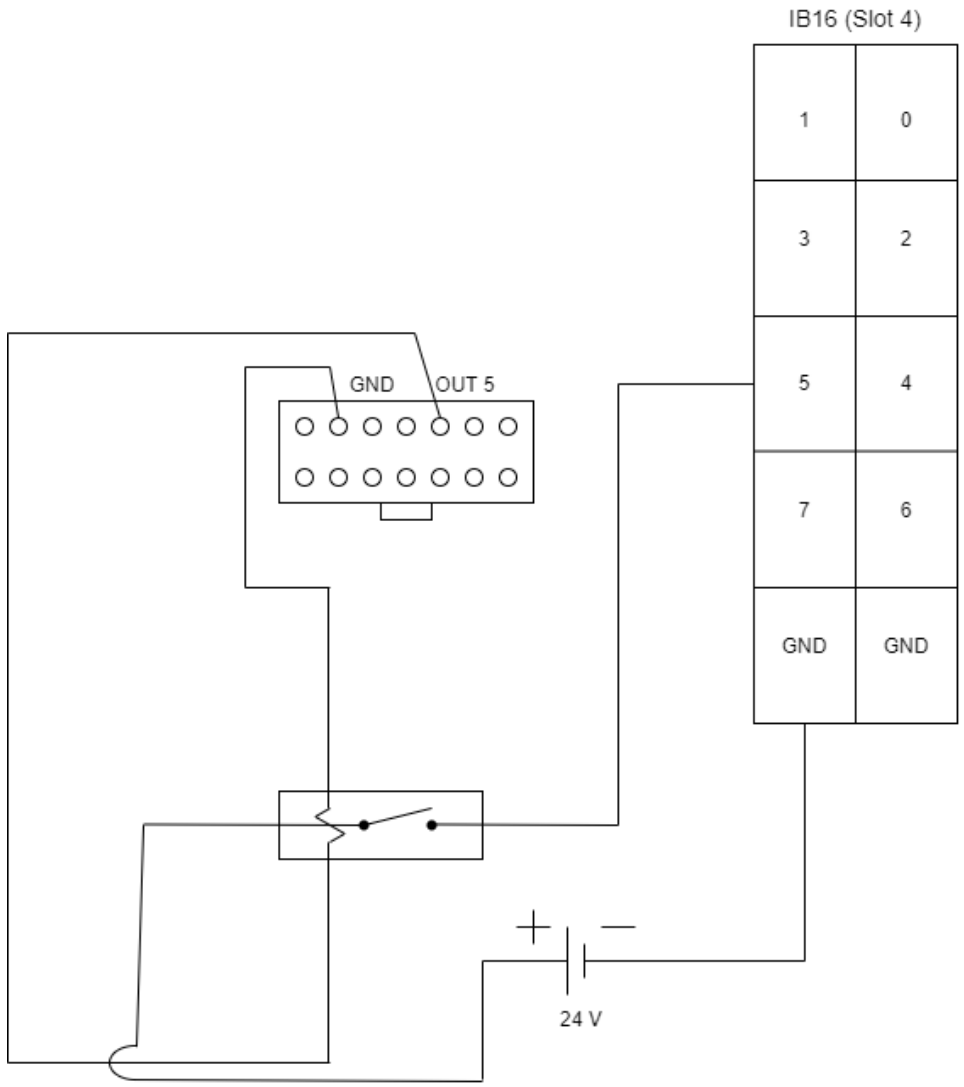
1.Process -

Robot Out 5 On --- send a signal to coil terminal --- contact terminal close --- IN5 of PLC (slot4) is trigged --- energize the OUT5 of PLC (slot9) --- light on for 2s

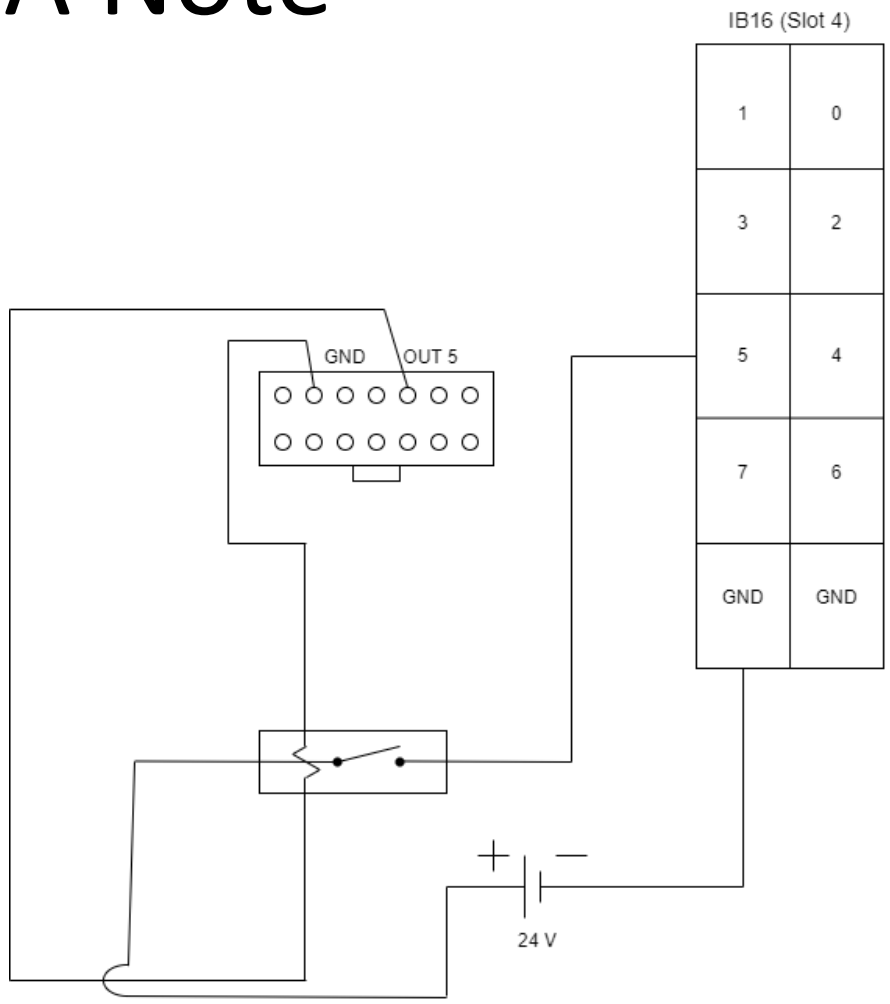
2.In your report:

Answers to the questions and

Drawing of the wire connection



TA Note



Process -

Out 5 On --- send a signal to coil terminal --- contact terminal close --- IN5 of PLC (slot4) is trigged --- energize the OUT5 of PLC (slot9) --- light on for 2s

Always Keep Warning:

Must confirm:

Slot4 input 5 (Number 6) is connect to 24V Positive

Slot4 GND 0 (number 10) connect to 24V Negative

Lab 10

- 1.
- OUT5, 1 - out 5 is on
- PAUSE 2 - keep on for 2s
- OUT5, 0 - out5 is off
- PAUSE 2 - keep off for 2s

(repeat)